

CHI-SQUARE TEST

- ❖ A chi-square test is a statistical test that is used to **compare observed and expected results**.
- ❖ A chi-square test is a statistical hypothesis test that surveys whether two categorical variables are independent in influencing the test statistic.
- ❖ It's used to compare observed results with expected results to determine if a difference is due to chance or a relationship between the variables.
- ❖ **The goal of this test is to identify whether a inequality between actual and predicted data is due to chance or to a link between the variables under consideration.**
- ❖ A chi-square test or comparable nonparametric test is required to test a hypothesis regarding the distribution of a categorical variable.
- ❖ Categorical variables, which indicate categories such as animals or countries, can be nominal or ordinal. They cannot have a normal distribution since they can only have a few particular values.

- ❖ It is used to calculate the difference between two categorical variables, which are:
 - ❖ As a result of chance
 - ❖ Because of the relationship
- ❖ Formula For Chi-Square Test

$$\chi^2_c = \frac{\sum (O_i - E_i)^2}{E_i}$$

Where

c = Degrees of freedom

O = Observed Value

E = Expected Value

- ❖ The degrees of freedom in a statistical calculation represent the number of variables that can vary in a calculation. The degrees of freedom can be calculated to ensure that chi-square tests are statistically valid. These tests are frequently used to compare observed data with data that would be expected to be obtained if a particular hypothesis were true.
- ❖ The Observed values are those you gather yourselves.
- ❖ The expected values are the frequencies expected, based on the null hypothesis.